

# Dr. Wei Guo

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## Academic Positions

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| <b>Hong Kong University of Science and Technology</b><br>Postdoctoral Fellow. Hosts: <a href="#">Prof. Khaled B. Letaief</a> and <a href="#">Prof. Jun Zhang</a> | Sept. 2023 – Present<br>Hong Kong SAR, China |
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## Education

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| <b>The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)</b><br>Ph.D., Computer and Information Engineering. Advisor: <a href="#">Prof. Chuan Huang</a> | Jul. 2023<br>Shenzhen, China |
| <b>University of Electronic Science and Technology of China (UESTC)</b><br>B.Eng., Communications Engineering                                                   | Jun. 2017<br>Chengdu, China  |

## Academic Visits

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| <b>Purple Mountain Laboratories</b><br>Visiting student.                                                                                                                          | Sept. 2021 – Nov. 2021<br>Nanjing, China |
| <b>The Future Network of Intelligence Institute (FNii), CUHK-Shenzhen</b><br>Visiting student. Advisors: <a href="#">Prof. Chuan Huang</a> and <a href="#">Prof. Shuguang Cui</a> | Jun. 2019 – Dec. 2019<br>Shenzhen, China |
| <b>Shenzhen Reserach Institute of Big Data (SRIBD)</b><br>Visiting student. Advisor: <a href="#">Prof. Shuguang Cui</a>                                                           | Feb. 2017 – Jun. 2017<br>Shenzhen, China |

## Current Research Interests

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Wireless Communications, Generative AI, Integrated Communications and AI

## Publications (✉ denotes corresponding author)

### Journals:

- Y. Wang, **W. Guo**<sup>✉</sup>, C. Huang<sup>✉</sup>, G. Chen, Y. Zhang, and S. Cui, “[Point Cloud Environment-Based Channel Knowledge Map Construction](#),” *IEEE Trans. Cogn. Commun. Netw.*, early access, 2026
- J. Tong, **W. Guo**, J. Shao, Q. Wu, Z. Li, Z. Lin, and J. Zhang<sup>✉</sup>, “[WirelessAgent: Large Language Model Agents for Intelligent Wireless Networks](#),” *China Commun.*, vol. 23, no. 3, pp. 265-285, Mar. 2026.
- T. Liao, **W. Guo**<sup>✉</sup>, H. He, S. Song, J. Zhang, and K. B. Letaief, “[Joint Beamforming and Antenna Position Optimization for Fluid Antenna Assisted MU-MIMO Networks](#),” *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 1209-1226, 2026.
- J. Shao, J. Tong, Q. Wu, **W. Guo**, Z. Li, Z. Lin, and J. Zhang<sup>✉</sup>, “[WirelessLLM: Empowering Large Language Models Towards Wireless Intelligence](#),” in *J. Commun. Inf. Netw. (JCIN)*, vol. 9, pp. 99-112, Jun. 2024.
- W. Guo**, C. Huang<sup>✉</sup>, X. Qin, L. Yang, and W. Zhang, “[Dynamic Clustering and Power Control for Two-Tier Wireless Federated Learning](#),” in *IEEE Trans. Wireless Commun. (TWC)*, vol. 23, no. 2, pp. 1356-1371, Feb. 2024.

Updated on: June 24, 2026

- W. Guo, R. Li, C. Huang<sup>✉</sup>, X. Qin, K. Shen, and W. Zhang, “Joint Device Selection and Power Control for Wireless Federated Learning,” in *IEEE J. Sel. Areas Commun. (JSAC)*, vol. 40, no. 8, pp. 2395-2410, Aug. 2022.
- W. Guo, H. Zhang<sup>✉</sup>, and C. Huang, “Energy Efficiency of Two-Way Communications under Various Duplex Modes,” in *IEEE Internet Things J. (IoT-J)*, vol. 8, no. 3, pp. 1921-1933, Feb. 2021.

#### Magazines:

- Y. Wang, C. Huang<sup>✉</sup>, S. Zhang, G. Chen, W. Guo<sup>✉</sup>, S. Lan, L. Xu, X. Cheng, X. Tang, and S. Cui, “Towards Precise Channel Knowledge Map: Exploiting Environmental Information from 2D Visuals to 3D Point Clouds,” *IEEE Commun. Mag.*, to appear, 2026.

#### Conference Proceedings:

- T. Liao, W. Guo, J. Qian, S. Song, J. Zhang, S. Jin, and K. B. Letaief, “Rethinking Mutual Coupling in Movable Antenna MIMO Systems,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, May. 2026.
- Y. Bian, W. Guo, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, “Multi-modal Data Driven Virtual Base Station Construction for Massive MIMO Beam Alignment,” in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026.
- E. Tang, W. Guo, H. He, S. Song, J. Zhang, and K. B. Letaief, “Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models,” in *Proc. IEEE Glob. Commun. Conf. (GlobeCom)*, Dec. 2025.
- T. Liao, W. Guo, H. He, S. Song, J. Zhang, and K. B. Letaief, “Fluid Antenna-Assisted MU-MIMO Systems with Decentralized Baseband Processing,” in *Proc. IEEE Glob. Commun. Conf. (GlobeCom)*, Dec. 2025.
- W. Guo, M. He, C. Huang, H. He, S. Song, J. Zhang, and K. B. Letaief, “Model-Driven Deep Learning for Distributed Detection with Binary Quantization,” in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2024.
- W. Guo, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, “Optimized Device Selection and Power Control for Wireless Federated Learning,” in *Proc. IEEE Glob. Commun. Conf. (GlobeCom)*, Dec. 2022.
- W. Guo, C. Huang, X. Qin, L. Yang, and W. Zhang, “Dynamic Clustering and Power Control for Two-Tier Over-the-Air Federated Learning,” in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Aug. 2022.
- W. Guo and C. Huang, “Energy Efficiency of Full-Duplex Two-Way Channels,” in *Proc. Glob. Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2019.

#### Patents:

- CN patent: “一种面向无线联邦学习的用户协同传输方法”, 黄川, 崔曙光, 郭玮. 专利申请号: 202211032064.0, 2022.

#### Preprints:

- E. Tang, W. Guo, H. He, S. Song, J. Zhang, K. B. Letaief, “A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS”, ([arXiv: 2605.29679](https://arxiv.org/abs/2605.29679)), 2026. (Submitted)
- T. Liao, W. Guo, J. Qian, Z. Wang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, “Rethinking Mutual Coupling in Movable Antenna MIMO Systems: Modeling and Optimization,” ([arXiv: 2604.26282](https://arxiv.org/abs/2604.26282)), 2026. (Submitted)

- B. Huang, Y. Wang, **W. Guo**, Z. Guo, D. Wu, R. Li, D. Liu, W. Lan, C. Huang, and S. Cui, “Building Low-Altitude Communication Networks: A Digital Twin-Based Optimization Framework,” ([arXiv: 2604.17781](#)), 2026. (Submitted)
- J. Tong, Z. Li, F. Liu, **W. Guo**, and J. Zhang, “WirelessAgent++: Automated Agentic Workflow Design and Benchmarking for Wireless Networks,” ([arXiv: 2603.00501](#)), 2026. (Submitted)

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## Teachings

**Teaching assistant** at CUHK-Shenzhen:

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| EIE4006 Performances Evaluation of Communication Networks            | Sum’ 2020-21 |
| CIE6126 Performances Analysis of Computer and Communication Networks | Sum’ 2020-21 |
| EIE3001 Signals and Systems                                          | Spr’ 2021    |

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## Awards

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| Best Paper Award at WCNC                                   | 2026 |
| Exemplary Reviewer of IEEE Wireless Communications Letters | 2020 |

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## Academic Services

### Conference Reviewers:

- IEEE Global Communications Conference (Globecom): 2019-2025
- IEEE/CIC International Conference on Communications in China (ICCC): 2019-2025
- IEEE Wireless Communications and Networking Conference (WCNC), 2024.
- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2024.

### Journal Reviewers:

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Communications
- IEEE Journal on Selected Areas in Information Theory
- IEEE Wireless Communications Letters
- IEEE Systems Journal
- IEEE Transactions on Machine Learning in Communications and Networking
- IEEE Internet of Things Magazine
- China Communications
- Journal of Communications and Information Networks
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## Presentations

2024 IEEE MeditCom, Marid, Spain, Oral presentation  
 2022 IEEE Globecom, Rio de Janeiro, Brazil, virtual presentation  
 2022 IEEE/CIC ICC, Foshan, China, Oral presentation  
 2019 IEEE GlobalSIP, Ottawa, Canada, Oral presentation

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